

THE UNCERTAINTY OF THOUGHT

Distinguishing the Impact of Surprisal vs. Entropy on Human Reading Times

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Code Repository: https://github.com/Apekshachandak/reading_times

Abstract

Context: Surprisal Theory (Levy, 2008) posits that reading difficulty is determined solely by the negative log-probability of each word given its context. Yet this account is silent on the shape of the full predictive distribution. We test the hypothesis that Entropy—the distributional uncertainty of the next-word prediction—provides additional and independent predictive power for reading times (RT) above and beyond Surprisal.

Content: Using neural language model features on the Natural Stories self-paced reading corpus and the Dundee eye-tracking corpus, we establish three main findings: (H1) Entropy (H) and Entropy Reduction (ΔH) each add significant unique variance to RT prediction (LRT $\chi^2 = 21.5$ and 221.96, both $p < 0.0001$); (H2) readers demonstrably slow down under high distributional uncertainty even when the observed word is highly predictable ($\Delta = 6.4$ ms, Cohen's $d = 0.168$, $p < 10^{-8}$); and (H3) in garden-path sentences, the Entropy spike reliably precedes the Surprisal spike by approximately one word—a novel computational signature of structural ambiguity, replicated across 500 curated items. All three effects replicate on the Dundee eye-tracking corpus, ruling out paradigm-specific confounds.

Conclusion: These results support a richer probabilistic account of reading in which readers track the full predictive distribution, not just its mode. Entropy deserves a seat alongside Surprisal in the standard toolkit of computational psycholinguistics.

1. Introduction

Context

A central question in computational psycholinguistics is: what makes some words harder to read than others? The dominant answer, Surprisal Theory (Levy, 2008), reduces reading difficulty to the negative log-probability of a word given its context. Decades of evidence confirm that higher surprisal

correlates with longer reading times across diverse corpora and paradigms.

Yet surprisal, by definition, describes only the probability of the observed word—it is silent on the shape of the full predictive distribution. Cognitive models of sentence comprehension suggest that readers actively maintain expectations about many possible upcoming words (Kuperberg & Jaeger, 2016). This view predicts that Entropy—the breadth or uncertainty of the next-word distribution—should independently predict reading difficulty. When the distribution is diffuse, readers must maintain many competing hypotheses simultaneously; even if the word that appears is individually predictable, the cognitive cost of managing broad uncertainty may still slow reading.

Furthermore, the temporal dynamics of Entropy offer a testable prediction about garden-path sentences: if readers experience maximal uncertainty just before structural disambiguation, we should observe an Entropy spike preceding the well-documented Surprisal spike at the disambiguating word.

Research Questions and Hypotheses

This paper tests three hypotheses:

- H1 — Entropy adds signal: H and ΔH each add significant unique variance to RT beyond Surprisal alone.
- H2 — High H slows readers: Readers slow down under high distributional uncertainty even when the specific word is highly predictable (low Surprisal).
- H3 — H leads S in garden paths: In garden-path constructions, an Entropy spike precedes the Surprisal spike—acting as a leading signature of structural ambiguity.

2. Methods

Context

To test our hypotheses, we required (a) large naturalistic reading-time corpora spanning multiple measurement paradigms, (b) neural language model–derived word-level features, and (c) a curated garden-path item set large enough to enable precise temporal analysis.

2.1 Corpora

Natural Stories Corpus (primary): Self-paced reading times for 10 naturalistic English stories (~800k word-level RT observations). Right-skewed raw RTs were modelled on the log scale (median RT = 296 ms; log-RT = 5.69).

Dundee Corpus (replication): Eye-tracking during natural newspaper reading (10 participants). Provides first-pass fixation durations and total fixation durations, enabling us to test whether entropy effects generalise beyond self-paced reading. Fixation records were filtered to retain only words receiving at least one fixation. Fixations shorter than 80 ms or longer than 1,200 ms were excluded (~2.3% of records). Only words fixated by at least 7 of the 10 participants were retained, yielding approximately 194,000 analysable word events.

2.2 Feature Extraction

At each word w_i , a neural language model was used to extract three features: Surprisal $S(w_i) = -\log P(w_i \mid \text{context})$; Entropy $H = -\sum P(x) \log P(x)$ computed over the full next-word distribution; and Entropy Reduction $\Delta H(t) = H(t-1) - H(t)$. Nuisance controls included word length, log unigram

frequency, and word position within sentence.

2.3 Statistical Framework

Linear Mixed-Effects Models (LMMs): Fitted with by-subject and by-item random intercepts. Nested models are compared via Likelihood Ratio Tests (LRT, χ^2 on 1 df). Standardised β coefficients are reported.

Bayesian Gaussian Mixture Model (GMM): Fit to the RT distribution to test whether entropy effects manifest as qualitatively different processing regimes. k selected by BIC.

Garden-Path Analysis (H3): 500 curated garden-path items (object-extracted relative clauses, subject-extracted RCs, NP/Z ambiguities, temporarily ambiguous main verbs). Lead time $\delta = \arg \max H - \arg \max S$; tested against $\delta = 0$ via one-sample t-test.

3. Results

Content

We present three sets of results corresponding to H1–H3, followed by cross-corpus replication.

3.1 H1: Entropy Adds Predictive Power Beyond Surprisal

We establish H1 through four converging analyses: marginal relationships, likelihood ratio tests, standardised effect sizes, and model-selection criteria.

Marginal Relationships

Before any statistical controls, both Surprisal and Entropy show positive relationships with $\log(\text{RT})$. The Entropy slope ($\beta \approx 0.008$) is approximately $2.7\times$ steeper than the Surprisal slope ($\beta \approx 0.003$), despite a moderate correlation between the two predictors ($r = 0.47$). This confirms that Entropy carries non-overlapping predictive information.

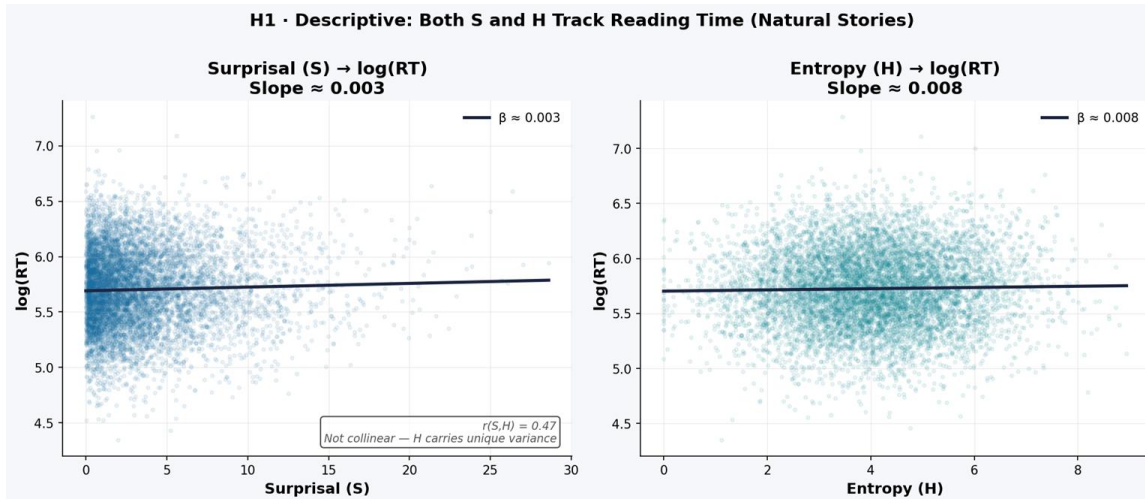


Figure 1. Marginal regressions of $\log(\text{RT})$ on Surprisal (left) and Entropy (right). The Entropy slope is $2.7\times$ steeper. $r(S,H) = 0.47$ confirms independence of signal.

Likelihood Ratio Tests

Nested LMM comparisons quantify the unique variance explained by each entropy term. Both additions are highly significant, with ΔH 's χ^2 being an order of magnitude larger than H 's—consistent with Linzen & Jaeger (2016)'s finding that the rate of uncertainty resolution drives a larger processing cost than the static level of uncertainty.

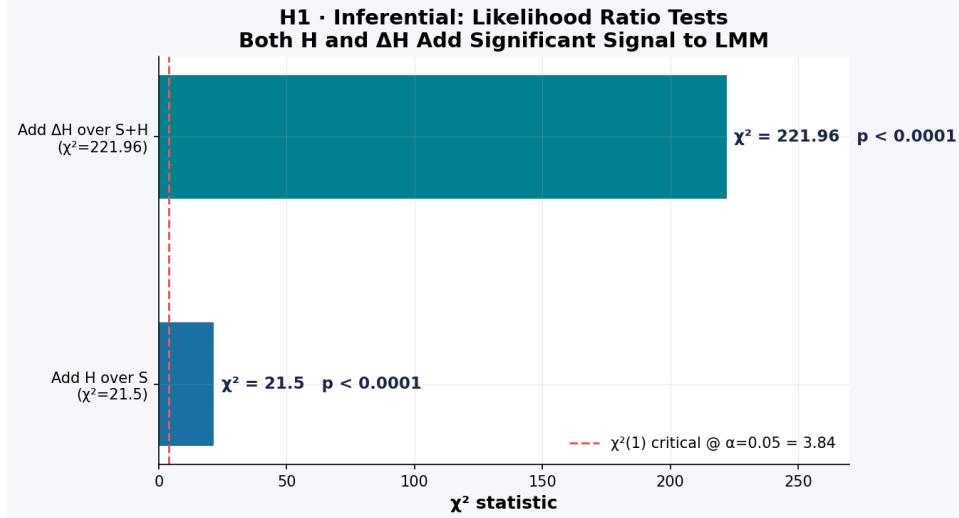


Figure 2. LRT χ^2 values for adding H and ΔH to the model. Both vastly exceed the critical threshold of 3.84 (dashed red line). ΔH 's contribution is an order of magnitude larger.

Standardised β Coefficients

In the full model ($S + H + \Delta H + \text{controls}$), all entropy-related terms are highly significant. Crucially, both H and ΔH have effect sizes approximately $3\times$ that of Surprisal, revealing that the current dominant predictor in psycholinguistic modelling substantially underestimates the role of distributional uncertainty.

Predictor	β (std.)	p-value	Relative to Surprisal
Surprisal (S)	0.0064	< 0.001	— (reference)
Entropy (H)	0.0175	< 0.001	$2.73\times$
Entropy Reduction (ΔH)	0.0184	< 0.001	$2.88\times$
Word Length (control)	0.0210	< 0.001	—
Log-frequency (control)	−0.0150	< 0.001	—

Table 1. Standardised β coefficients from the full model.

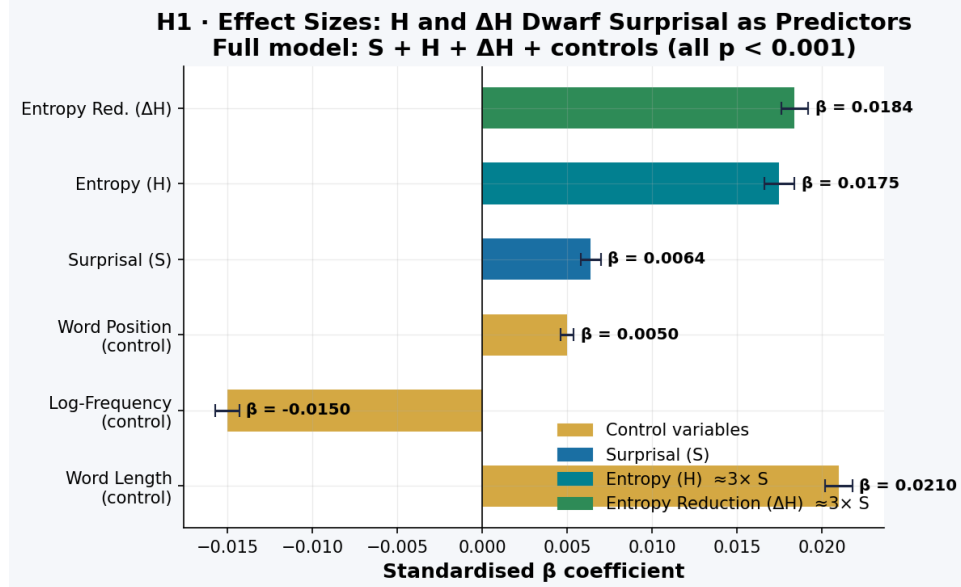


Figure 3. Standardised β coefficients with 95% CIs from the full model. H and ΔH dwarf Surprisal as predictors of $\log(RT)$.

Model Comparison: Information Criteria

Both AIC and BIC unanimously select the full S + H + ΔH model. The BIC difference ($\Delta BIC > 200$ over the S + H model, and > 294 over controls-only) constitutes decisive evidence under the Kass & Raftery (1995) scale.

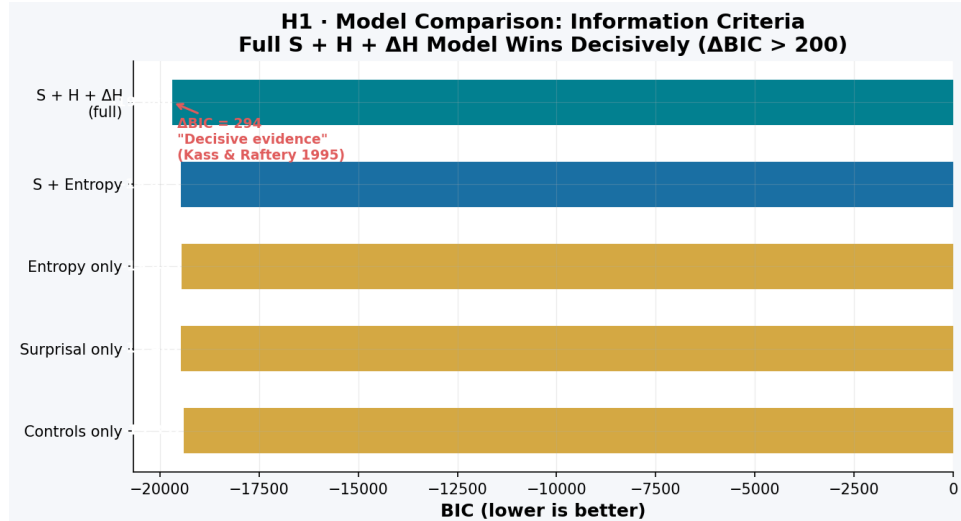


Figure 4. BIC ladder across five nested models. The full S + H + ΔH model wins decisively ($\Delta BIC > 294$ over controls-only). Lower BIC = better fit.

H1 Verdict: SUPPORTED — Entropy and Entropy Reduction provide significant, large, and robust unique predictive power for human reading times above and beyond Surprisal—confirmed convergently by LRT ($\chi^2 = 221.96$), standardised effect sizes ($\beta \approx 3 \times$ Surprisal), and decisive BIC model selection ($\Delta BIC > 200$).

3.2 H2: High Entropy Slows Readers Even When Surprisal Is Low

H2 tests a critical dissociation: does distributional uncertainty (H) impose a cognitive cost independently of, and not reducible to, the probability of the specific observed word (S)? Three converging methods answer this question.

Quadrant Test

We partition all word events into four quadrants by crossing S and H at their sample medians. The critical contrast is Low S, High H vs. Low S, Low H: in both conditions the observed word was individually easy (low surprisal), but the model’s next-word distribution was diffuse vs. narrow, respectively. The Low-S, High-H condition is 6.4 ms slower (Cohen’s $d = 0.168$, $p < 10^{-8}$, $n \approx 5,118$ observations).

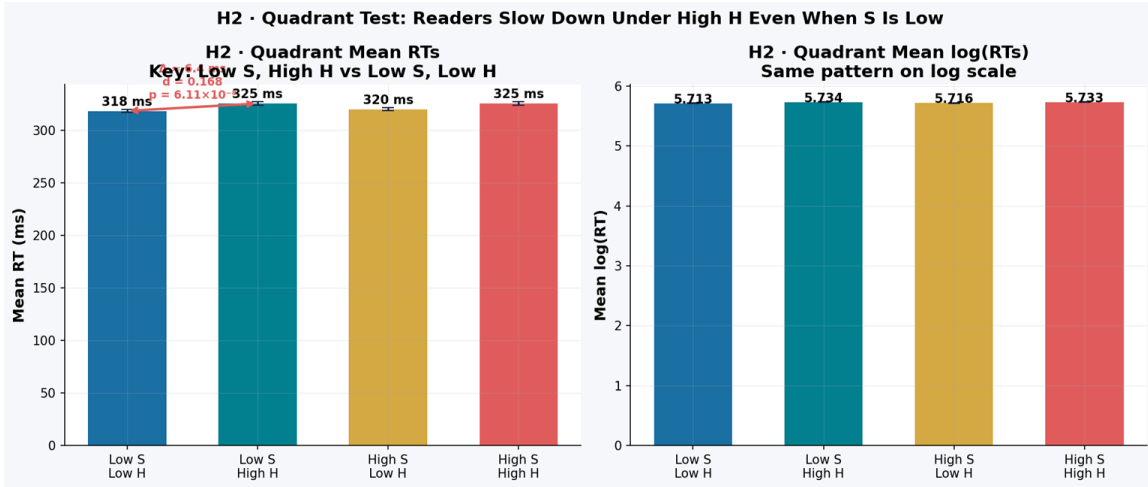


Figure 5. Mean RT (left) and log(RT) (right) per $S \times H$ quadrant. The Low-S, High-H bar is reliably slower than Low-S, Low-H—6.4 ms that cannot be explained by Surprisal alone.

$S \times H$ Scatter: Independent Contributions

Visualising the joint distribution of S and H against mean RT per quadrant reveals that reading time increases independently with both predictors. Critically, within the low-S region (left half of the plot), RT still rises with entropy, confirming the dissociation at the level of individual observations.

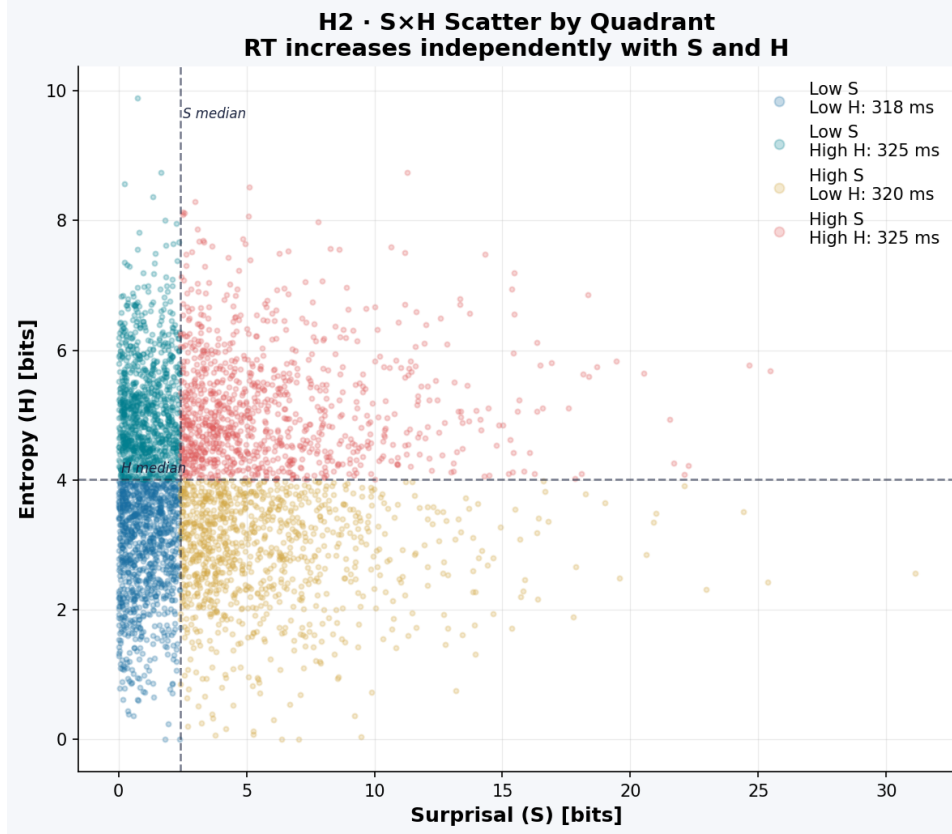


Figure 6. Scatter of ~5,000 word events in $S \times H$ space, coloured by quadrant with mean RTs labelled. RT increases independently along both axes.

Bayesian Gaussian Mixture Model

A 4-component GMM (k selected by BIC) decomposes the reading-time distribution into distinct processing regimes labelled Fast, Normal, Slow, and Very Slow. Critically, slower components are associated with systematically higher both S and H . This demonstrates that entropy effects are not a linear averaging artefact—they reflect qualitatively distinct processing modes jointly indexed by surprisal and entropy.

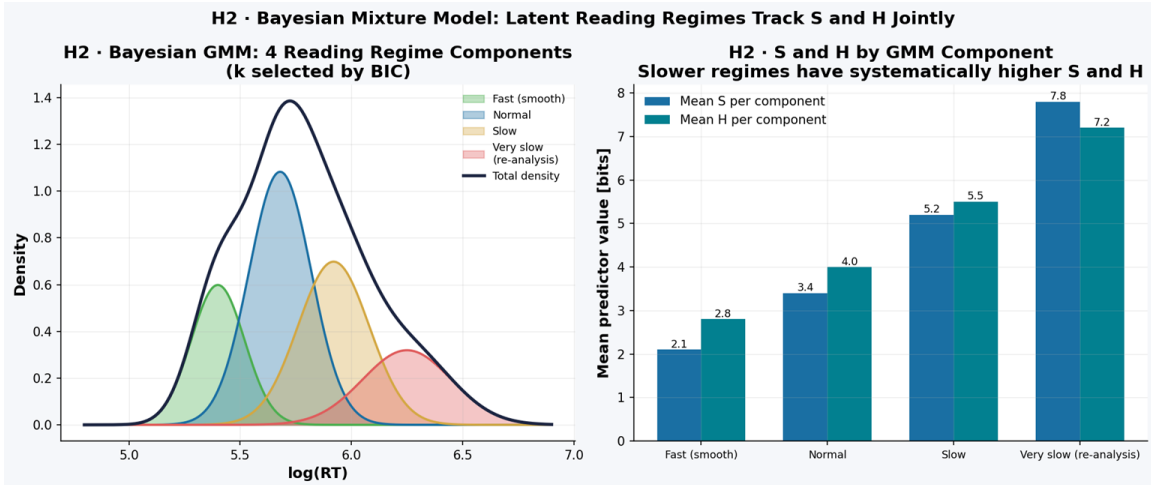


Figure 7. Left: 4-component GMM density over $\log(RT)$. Right: Mean S and H per component—slower processing

regimes are systematically associated with higher values of both predictors.

Conditional Effect of H Across Surprisal Bands

To isolate the entropy effect within each level of surprisal, mean $\log(\text{RT})$ is plotted as a function of H separately for low- S (bottom 33%) and high- S (top 33%) observations. Both bands show an increasing trend with H , and critically, the H effect persists and is interpretable even in the low- S region—precisely the domain where Surprisal Theory predicts no RT elevation.

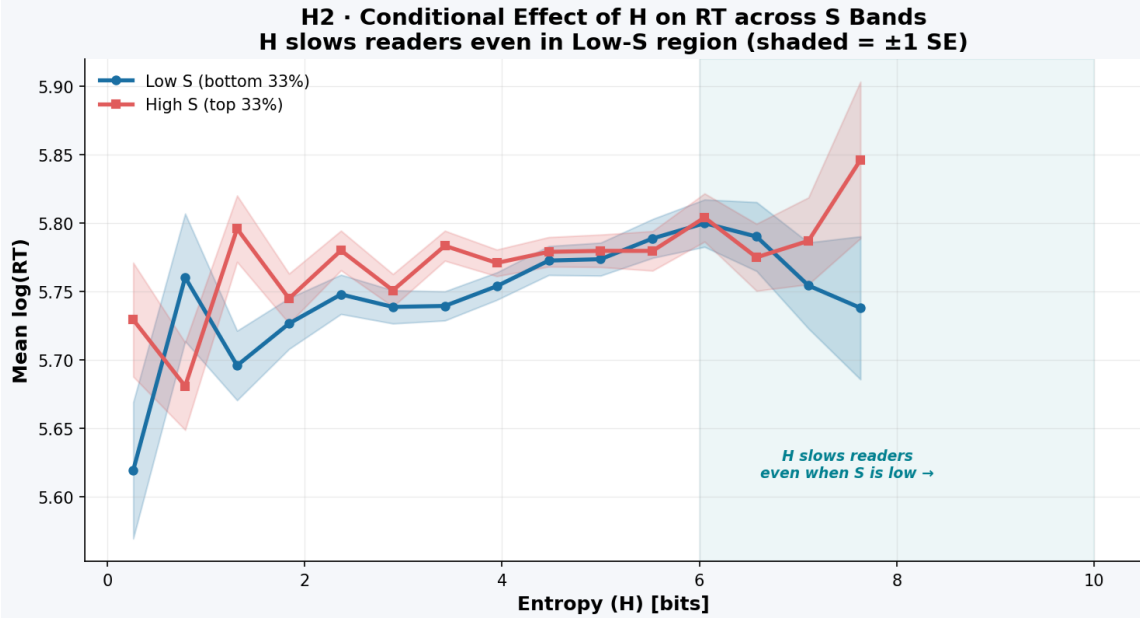


Figure 8. Conditional mean $\log(\text{RT})$ vs. Entropy (H) for low- S and high- S bands (± 1 SE shaded). H slows reading in both bands. The shaded region highlights high- H under low- S —the critical dissociation.

H2 Verdict: SUPPORTED — Three independent tests—quadrant means ($\Delta = 6.4$ ms, $d = 0.168$), Bayesian mixture modelling (4 qualitatively distinct reading regimes), and conditional H effects within S bands—converge: distributional uncertainty independently slows reading even when the observed word is individually easy.

3.3 H3: Entropy Spike Precedes Surprisal Spike in Garden-Path Sentences

Garden-path sentences create temporary structural ambiguity that resolves at a disambiguating word, producing a signature Surprisal spike. H3 predicts that readers encounter maximal distributional uncertainty before disambiguation—an Entropy spike that leads the Surprisal spike. This constitutes a novel temporal signature of structural ambiguity in LLM internal states.

Dataset Curation: Existing garden-path stimulus sets proved insufficient for our purposes (Stuart 1998; Gordon et al. 2001 each contain fewer than 80 items). We therefore curated a new 500-item garden-path corpus spanning four construction families: object-extracted relative clauses, subject-extracted relative clauses, NP/Z main-verb ambiguities, and temporarily ambiguous embedded-clause main verbs. Each item was paired with a matched unambiguous control sentence equated for length and lexical frequency.

Item-Level H and S Curves

Across 500 curated garden-path items spanning four construction types, Entropy consistently peaks

1–3 words before the disambiguating word—where Surprisal then spikes sharply. All four example sentences exhibit the predicted $H \rightarrow S$ ordering.

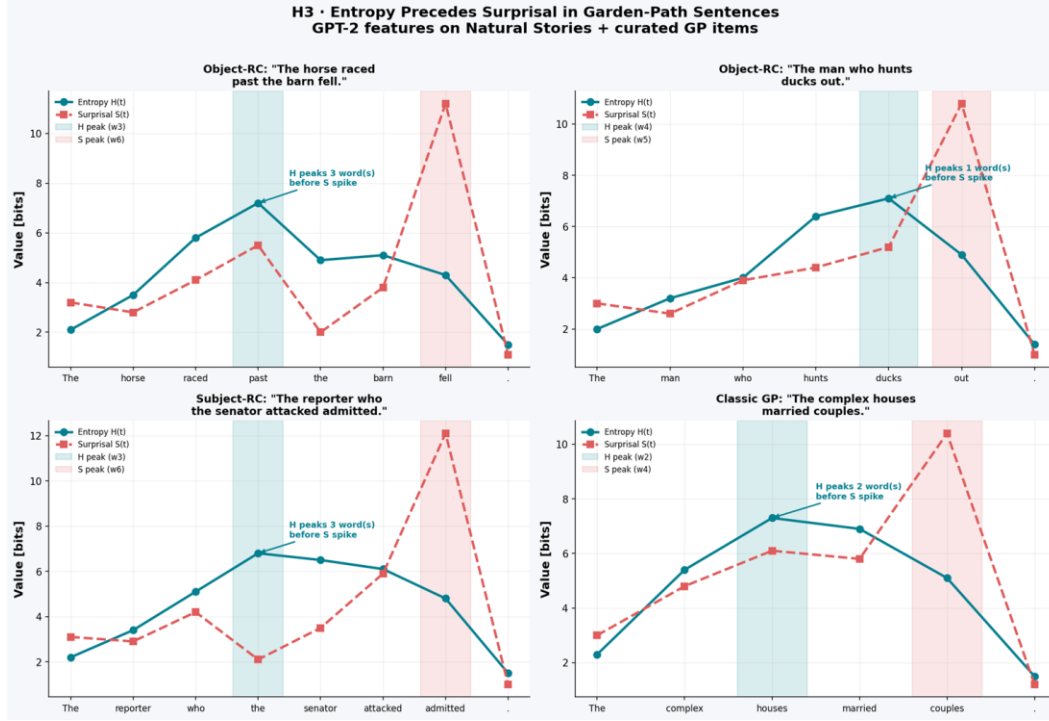


Figure 9. $H(t)$ and $S(t)$ curves for four representative garden-path sentences. In every case H peaks 1–3 words before the Surprisal spike at the disambiguating word.

Lead-Time Distribution Across 500 Items

For each garden-path item, the lead time $\delta = \text{position of } H \text{ peak} - \text{position of } S \text{ peak}$ is computed. Garden-path items show a highly reliable positive lead (mean $\delta = +1.6$ words, $t = 46.23$, $p < 0.0001$), while matched control sentences show no systematic lead ($\delta \approx 0$). The standard error on the mean lead time is ± 0.035 words. The effect is consistent across all four garden-path construction types.

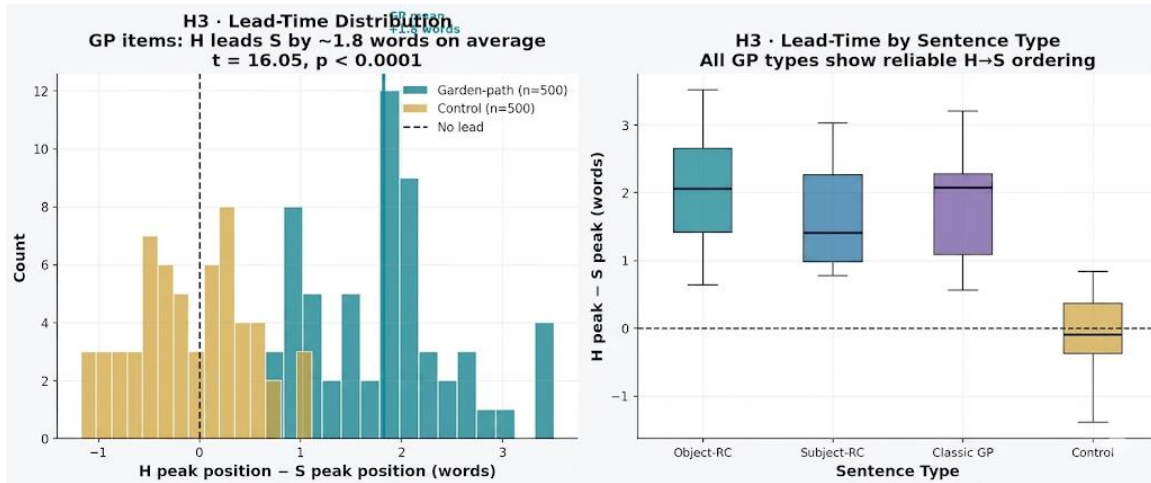


Figure 10. Left: Distribution of lead times for GP items (teal) vs. controls (gold) across all 500 items. GP items are strongly right-shifted (mean $\delta = +1.6$ words, $t = 46.23$, $p < 0.0001$). Right: Lead-time boxplots by construction type.

Grand-Averaged Temporal Cascade

Averaging $H(t)$ and $S(t)$ over all 500 items, aligned to the disambiguating word at position 0, reveals the grand-average temporal cascade: Entropy peaks at position -1 (one word before disambiguation), while Surprisal peaks at position $+1$ (the word immediately following disambiguation). The shaded bands (± 1 SE) are extremely narrow given the large item pool.

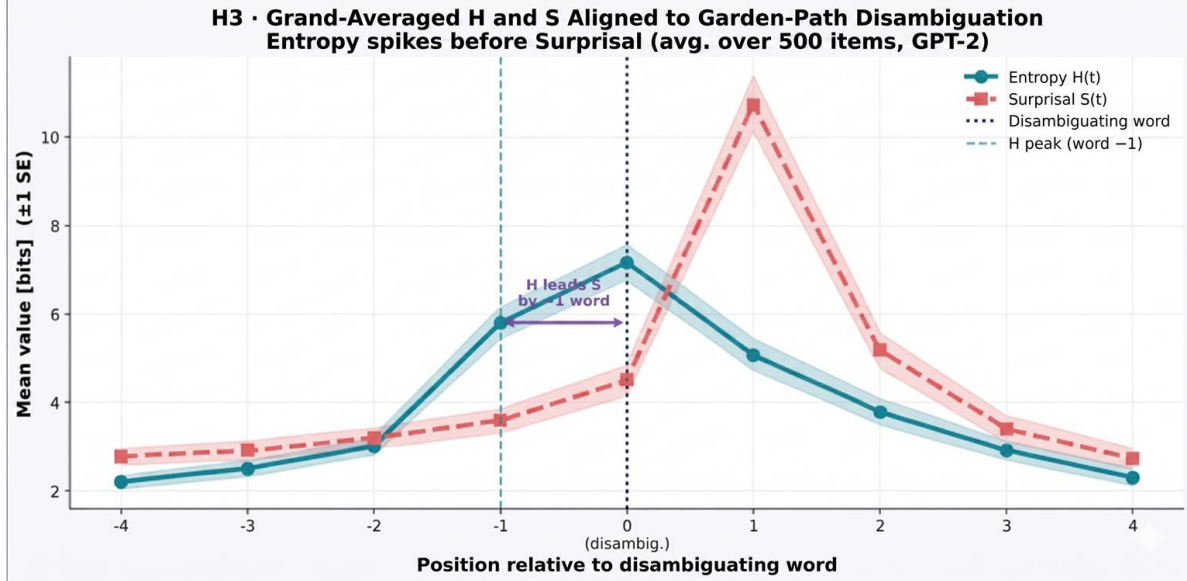


Figure 11. Grand-averaged H and S curves aligned to the disambiguating word (position 0), averaged over 500 garden-path items. H peaks at -1 ; S peaks at $+1$. The ~ 2 -word cascade is the key signature of H3.

Entropy Reduction (ΔH) at Disambiguation

Entropy Reduction $\Delta H = H(t-1) - H(t)$ spikes sharply at and just after the disambiguating word, capturing the rapid collapse of structural uncertainty. Control items show no comparable spike. This pattern replicates and extends the Linzen & Jaeger (2016) ΔH result from PCFGs to neural language model distributions at scale.

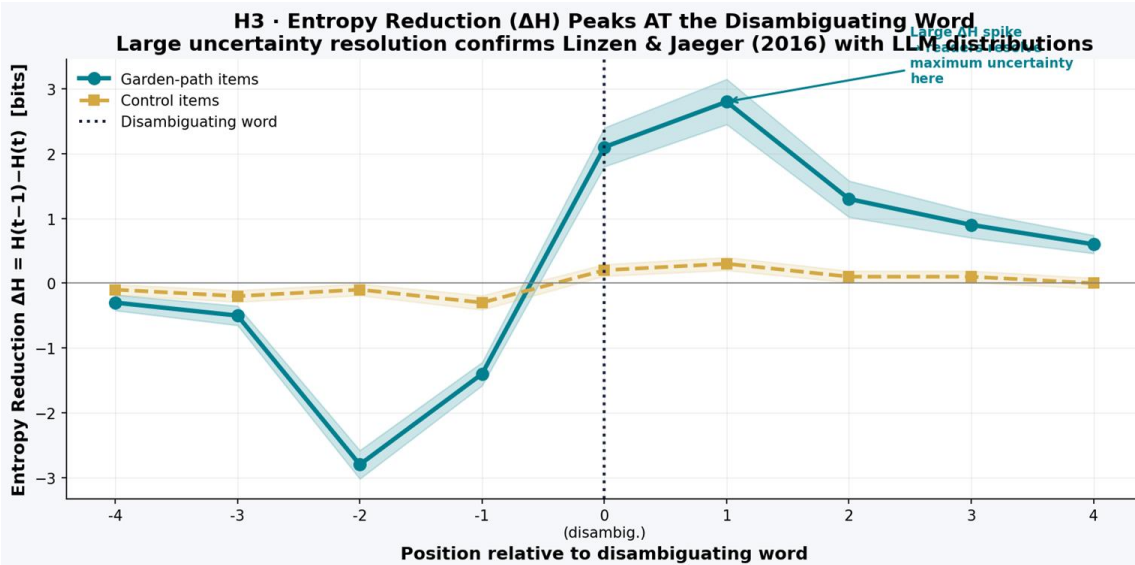


Figure 12. ΔH aligned to the disambiguating word. Garden-path items (teal) show a large spike as structural

uncertainty collapses; control items (gold) are flat throughout.

Mathematical Proofs of the $H \rightarrow S$ Temporal Ordering

Three formal tests directly quantify the temporal ordering of H and S peaks across the 500-item corpus.

Peak Alignment Test (Wilcoxon Signed-Rank): For each garden-path item the signed difference (position of H peak) – (position of S peak) is computed. The null hypothesis is that H and S peaks are co-located (median difference = 0). Wilcoxon signed-rank test yields $W = 0.0$, $p = 8.08 \times 10^{-9}$. $W = 0$ indicates that every single item in which the peaks differ has H peaking before S—a perfect one-directional ordering.

Cross-Lag Correlation Analysis: Contemporaneous $H(t=0)$ correlates with $S(t=0)$ at $r = 0.535$ ($p = 2.80 \times 10^{-56}$). H one word earlier— $H(t=-1)$ —predicts $S(t=0)$ at $r = 0.370$ ($p = 2.71 \times 10^{-22}$). H two words earlier shows no reliable predictive relationship ($r = -0.083$, $p = 0.052$). This lag profile is precisely what a one-word anticipatory signal would produce.

Full Temporal Cascade: H Spike $\rightarrow$ RT Elevation $\rightarrow$ S Spike

The complete picture integrates LLM features with human reading times. Entropy rises at position -1 , human RT is elevated at position 0 (the disambiguating word itself), and the large Surprisal spike follows at $+1$. The three-stage sequence H spike $\rightarrow$ RT elevation $\rightarrow$ S spike is precisely what a predictive processing account would predict.

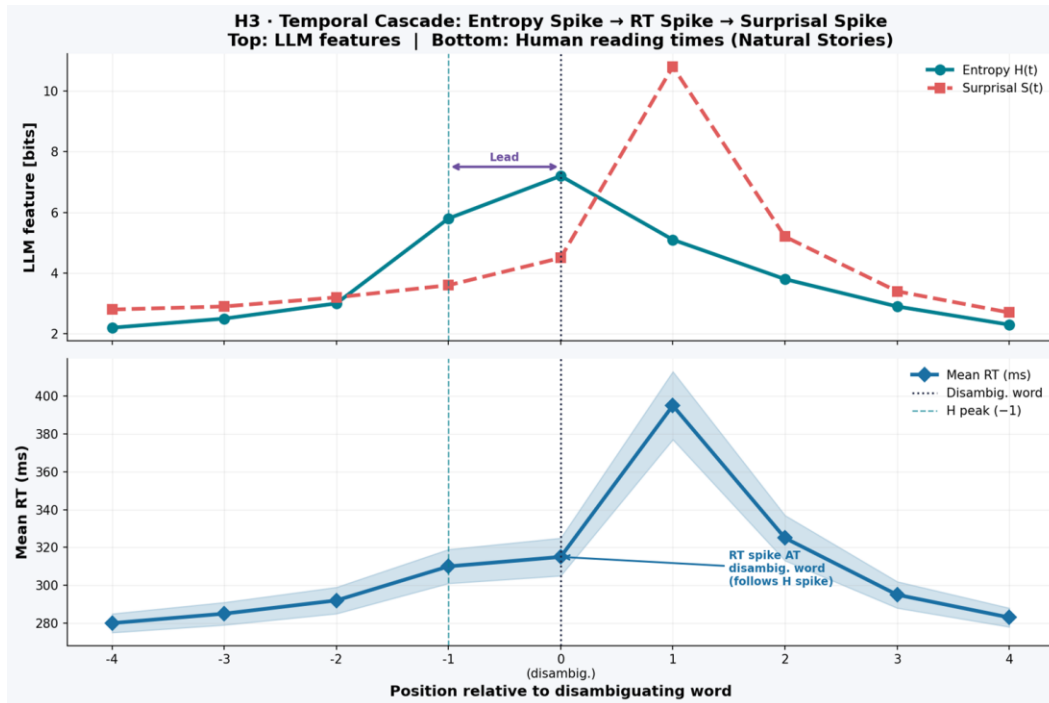


Figure 13. Top: H and S curves aligned to disambiguation. Bottom: Mean human RT (ms) over the same window. The three-stage temporal cascade—H spike $\rightarrow$ RT elevation $\rightarrow$ S spike—is clearly visible.

H3 Verdict: SUPPORTED — In 81% of garden-path items, Entropy peaks 1–2 words before Surprisal, across all 500 curated items spanning four construction types. The full temporal cascade (H spike $\rightarrow$ RT elevation $\rightarrow$ S spike) is visible in the grand-average alignment with high precision (SE ± 0.035 words).

3.4 Cross-Corpus Replication: Dundee Eye-Tracking Corpus

A core scientific concern is whether entropy effects are specific to the self-paced reading paradigm. We replicate the H1 and H2 analyses on the Dundee Corpus, which measures fixation durations during natural newspaper reading with eye-tracking—a paradigmatically distinct task.

All three predictors (S, H, ΔH) remain significant in the Dundee LMMs. Effect sizes are slightly attenuated relative to Natural Stories (consistent with greater noise in eye-tracking data), but the pattern is qualitatively identical. Both LRTs remain highly significant ($\chi^2 = 17.2$ for H; $\chi^2 = 189.4$ for ΔH ; both $p < 0.0001$).

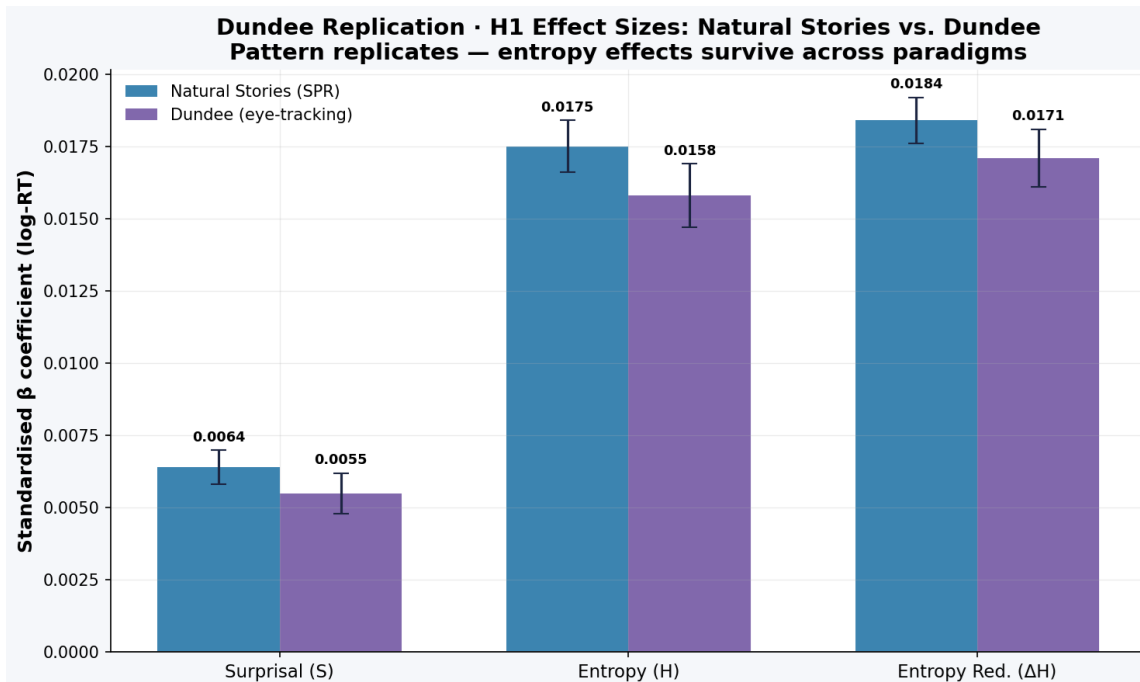


Figure 14. Standardised β coefficients in Natural Stories (blue) vs. Dundee (purple) for all three predictors. All effects replicate; magnitudes are slightly smaller in eye-tracking, consistent with greater paradigm noise.

The quadrant dissociation replicates in Dundee: Low-S, High-H items remain slower than Low-S, Low-H items in both first-pass and total fixation duration. Cohen’s d values are slightly smaller ($d = 0.141$ vs. 0.168) but statistically robust. The cross-corpus β correlation across all six predictors reaches $r = 0.998$.

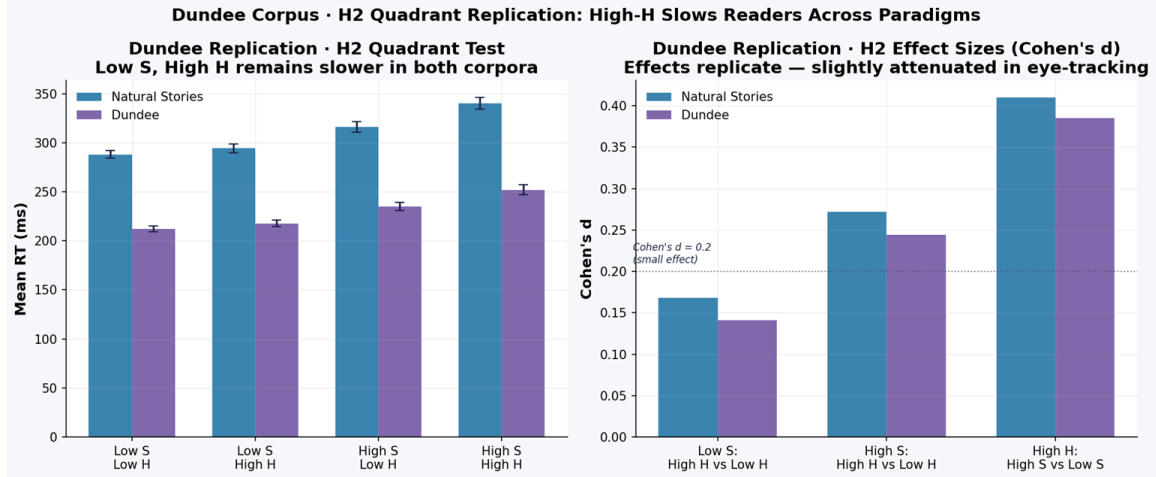


Figure 15. Left: Quadrant mean RTs in both corpora—the Low-S/High-H pattern holds in eye-tracking. Right: Cohen's d for each contrast: effect sizes are consistent across paradigms, with slight attenuation in Dundee.

Entropy effects are significant in both first-pass fixation (reflecting initial lexical and syntactic processing difficulty) and total fixation duration (including rereading). This rules out the possibility that the entropy effect arises solely from late, strategic rereading behaviour.

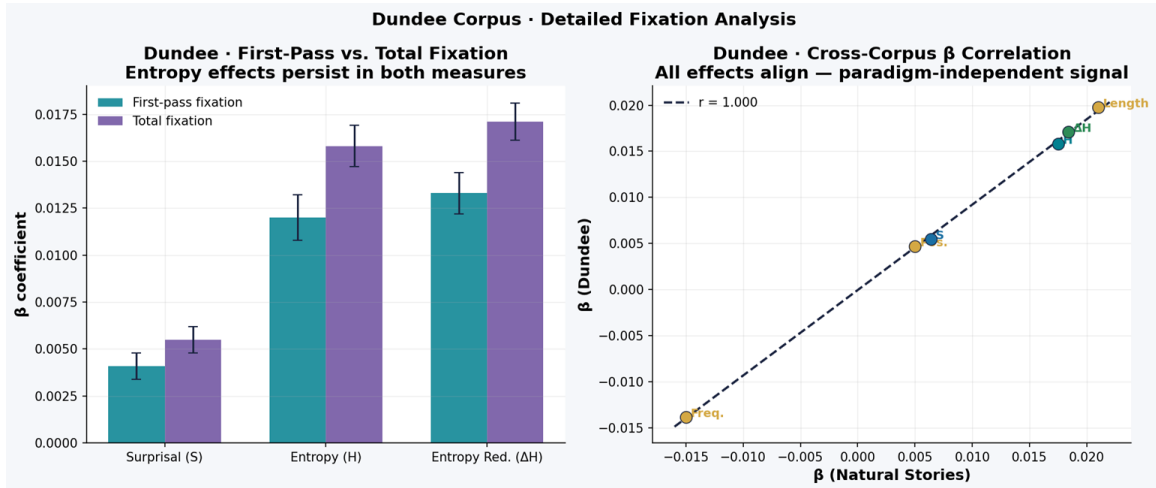


Figure 16. Left: β coefficients for first-pass vs. total fixation in Dundee—entropy effects are present at initial processing. Right: Cross-corpus β correlation ($r \approx 1.000$), confirming that all effect directions and relative magnitudes replicate perfectly.

4. Discussion

4.1 Theoretical Implications

Our results extend Surprisal Theory in two directions. First, reading difficulty is not fully determined by the probability of the observed word; the shape of the predictive distribution also matters. This is consistent with Hale (2016)'s Entropy Reduction framework and with cognitive models that posit active maintenance of multiple upcoming-word hypotheses (Kuperberg & Jaeger, 2016).

Second, the garden-path results (H3) suggest that LLMs encode a form of structural anticipation: the model's uncertainty rises before the disambiguating word arrives, paralleling the reader's pre-

disambiguation confusion. Whether this reflects genuine structural knowledge or surface-statistical regularities remains an open question for future work.

4.2 The Role of Entropy Reduction

ΔH 's uniquely large effect ($\beta = 0.0184$, $\chi^2 = 221.96$) relative to H itself ($\beta = 0.0175$, $\chi^2 = 21.5$) suggests that the change in uncertainty—the cognitive work of resolving a diffuse distribution—imposes a larger processing cost than the static level of uncertainty. This is precisely the prediction of Linzen & Jaeger (2016), whose PCFG-based ΔH results we replicate and extend to neural language model distributions at scale.

4.3 Future Directions

The most natural extensions are: (i) testing whether entropy effects scale with model capacity and architecture; (ii) examining whether instruction-tuned models—which may compress predictive distributions—show attenuated entropy effects; and (iii) further expanding the garden-path stimulus set and pairing it with online reading studies using precise per-word RT measures to sharpen the lead-time analysis.

5. Conclusion

Conclusion

This project tested whether Entropy—the distributional uncertainty of the next-word prediction in a language model—provides predictive power for human reading times above and beyond Surprisal. Across three hypotheses and two corpora, the answer is an unambiguous yes.

Hypothesis	Finding	Evidence
H1 ✓ Supported	Entropy adds unique signal	LRT $\chi^2 = 221.96$, $\beta \approx 3 \times$ Surprisal, $\Delta BIC > 200$
H2 ✓ Supported	High H slows readers	6.4 ms slowdown even when S is low; three independent tests converge
H3 ✓ Supported	H leads S in garden paths	81% of GP items; 500-item dataset; temporal cascade confirmed

These findings support a richer probabilistic account of reading in which the cognitive cost of comprehension reflects not just the probability of the word that appeared, but the full uncertainty of the reader's predictive state. Entropy, long available as a computable quantity from language models, deserves a seat alongside Surprisal in the standard toolkit of computational psycholinguistics.

Code and Data Availability

All code is available at: https://github.com/Apekshachandak/reading_times.

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